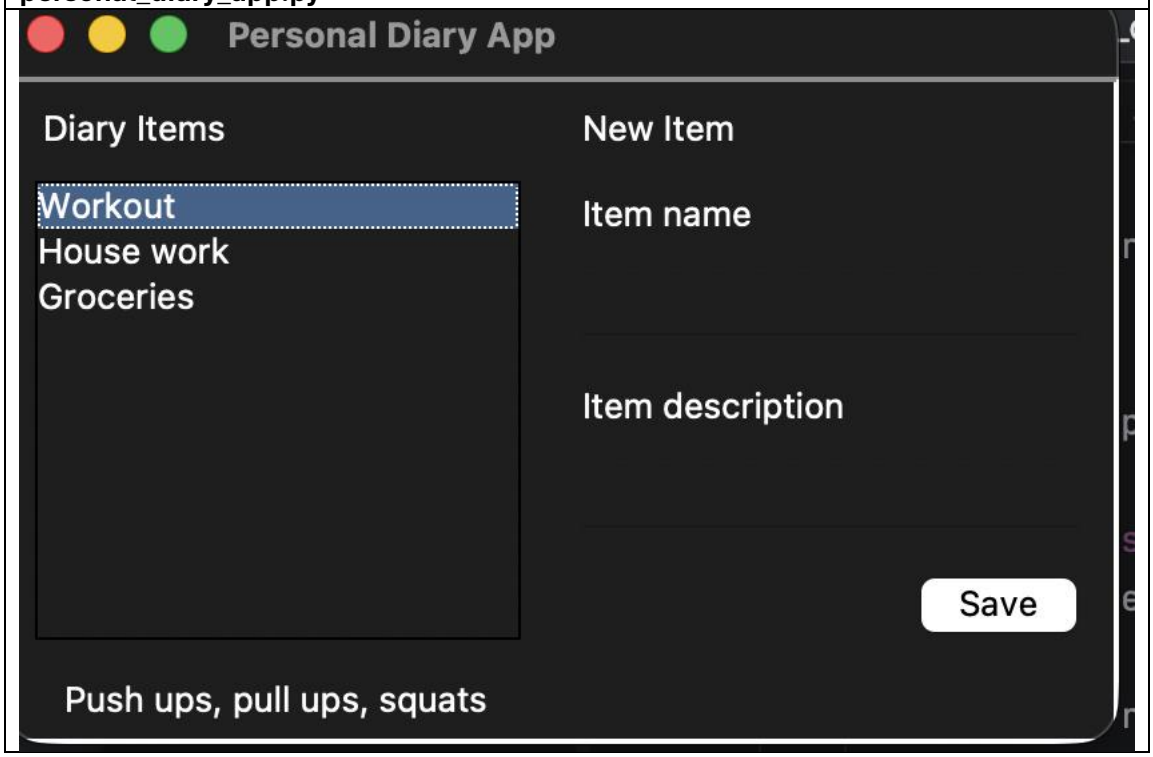
Week 4 Tutorial & Computer Lab

Assignment 2 Case Studies with Programming Problems

Case Study 1: Python Tkinter and Pillow packages

Task 1: Intro to Tkinter

personal_diary_app.py

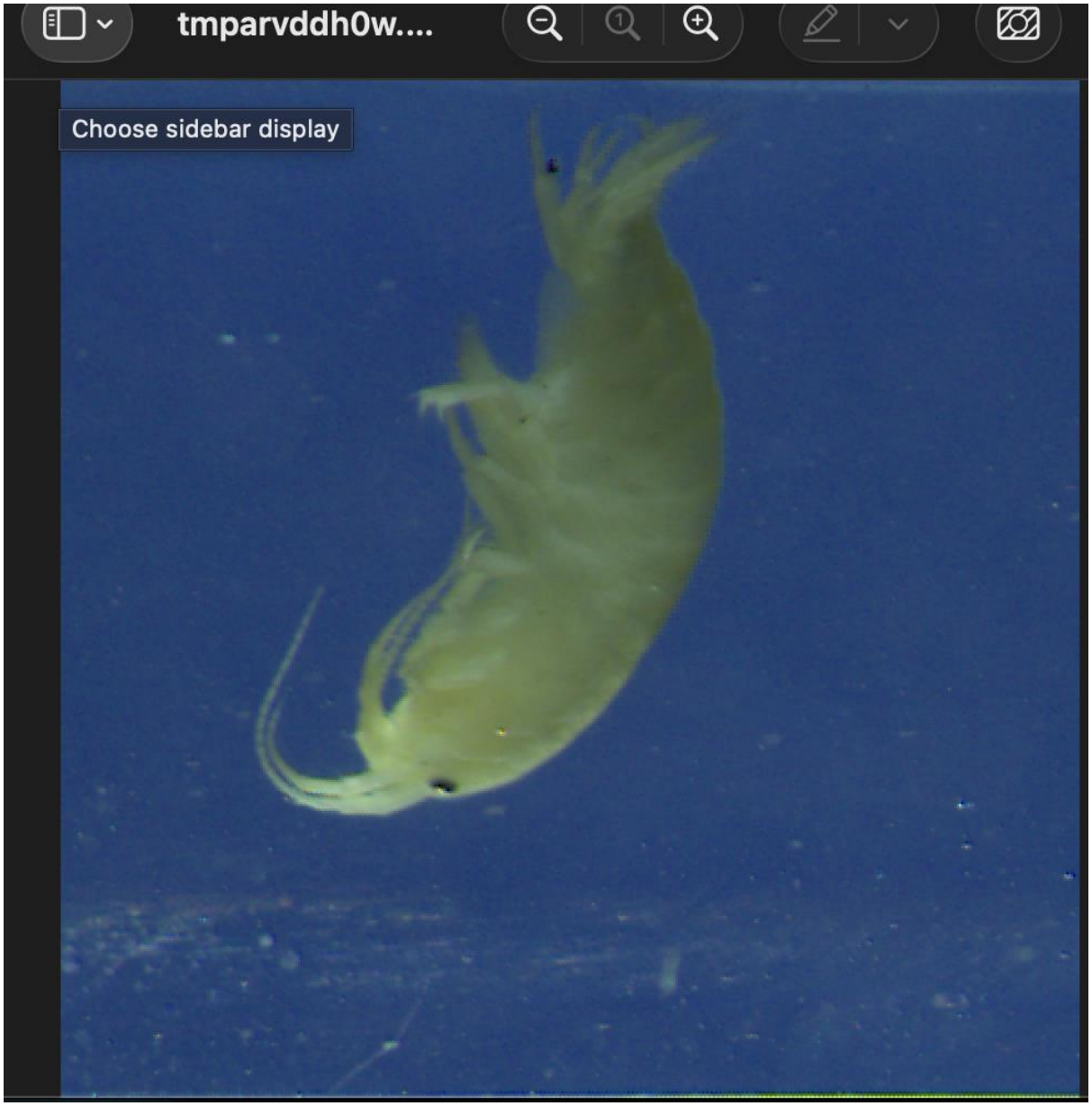


Task 2: Intro to Pillow

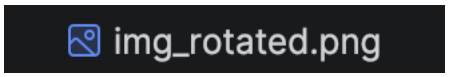
Basic Info Output

1- Insect 1

```
/usr/local/bin/python3.14 /Users/ali/Week 4 Tutorials & Computer Lab - Python Tkinter and Pillow package /intro.py
(464, 464)
Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.PNG
PNG
```



- Creates New imagine in Folder



2 – Cat Image

```
/usr/local/bin/python3.14 /Users/ali/Week 4 Tutorials & Computer Lab - Python Tkinter and Pillow package /intro.py
(1920, 2879)
Assets/Cat.JPG
JPEG
```

Creates New photo

cat_rotated.png

Task 3: Manipulating

Image 1: 1_Gammar 2021_1_2021_06_03-13-19-23-898.PNG

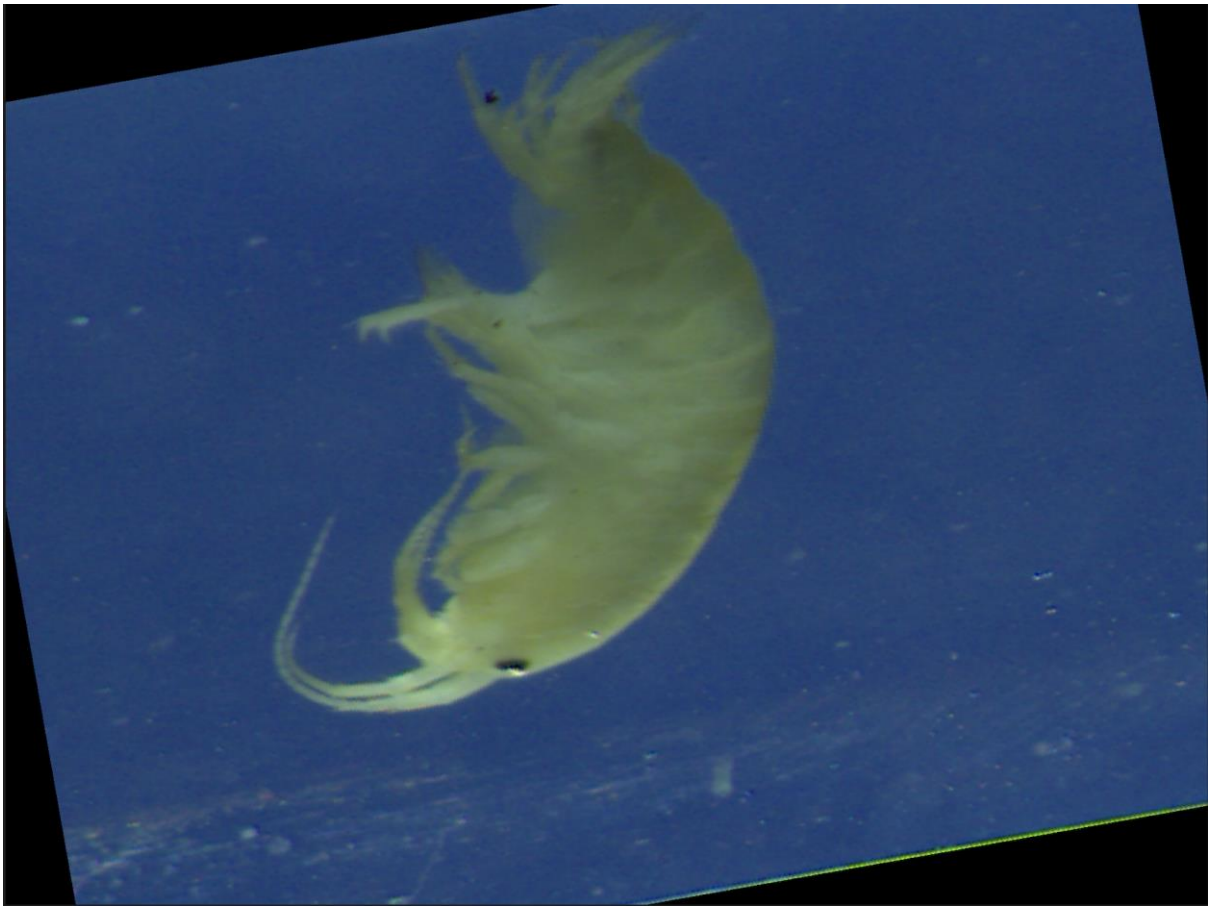
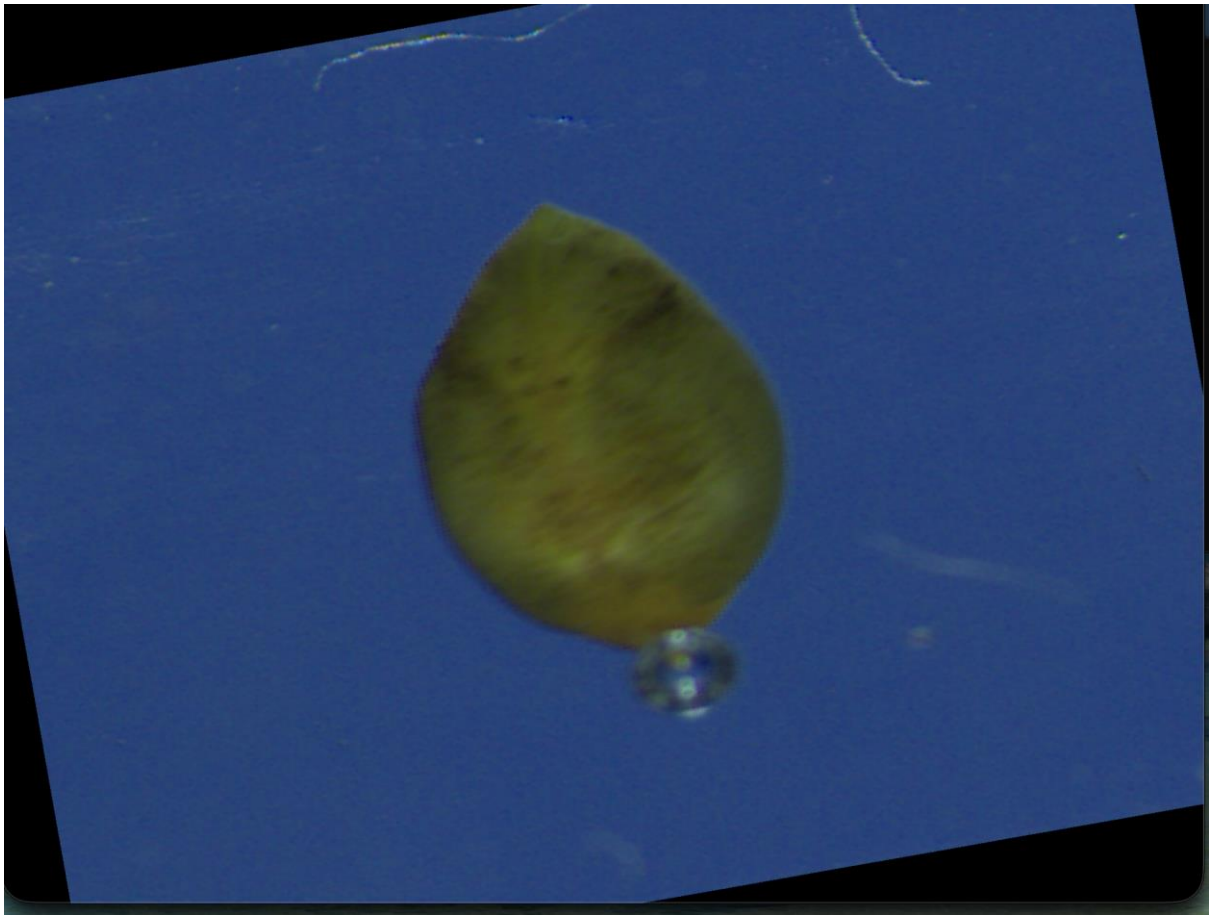


Image 2: 1_Limniu 2021_1_2021_06_02-13-08-37-958.PNG

Image 3: 2_Sphaer 2021_1_2021_06_03-11-22-54-852.PNG



Task 4 : Colour change

Image 1: 2_Sphaer 2021_1_2021_06_03-11-22-54-894.PNG

Colour (36, 73, 124)

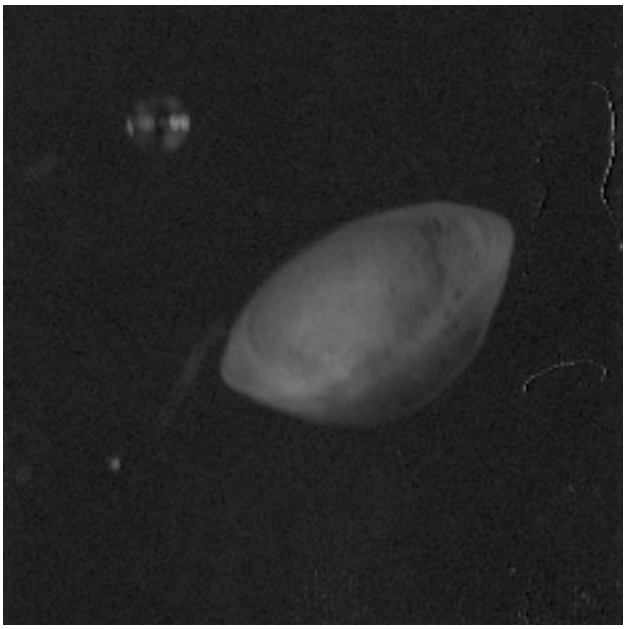


Image 2: 1_Limniu 2021_1_2021_06_02-13-08-37-941.PNG

Colour: (43, 88, 118)

Image 3: Dog.jpeg

Colour: (186, 170, 155)

Task 5 : Image enhancement

Image 1: Dog.jpg

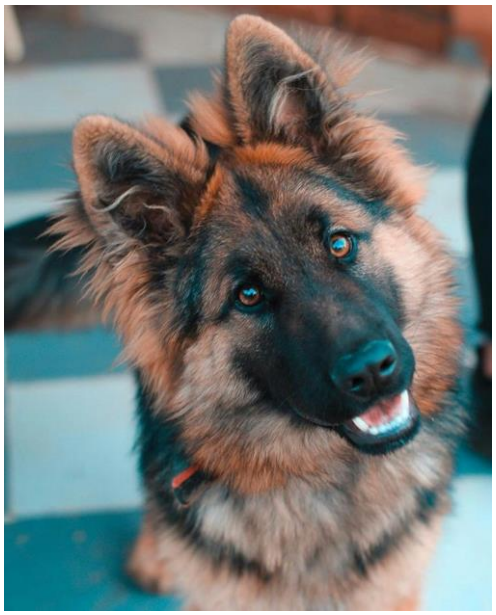
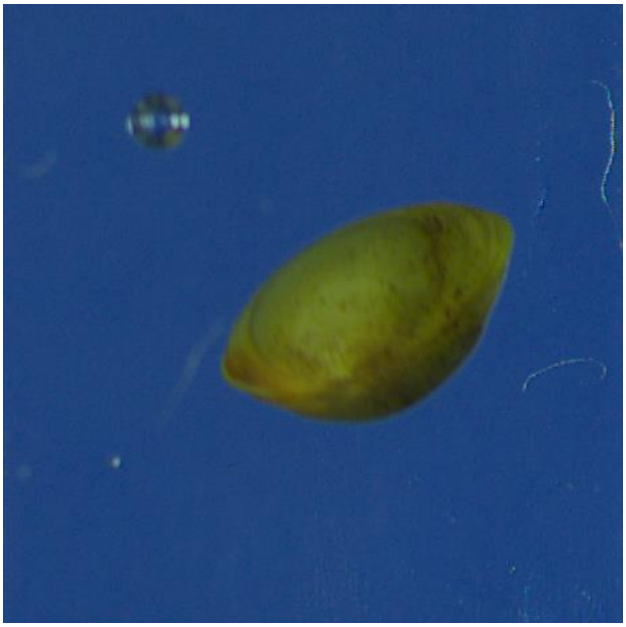


Image 2: 1_Gammar 2021_1_2021_06_03-13-19-23-856.PNG

Image 3 : 2_Sphaer 2021_1_2021_06_03-11-22-54-894.PNG



Task 6: Image filtering

Image 1 : Dog.jpg

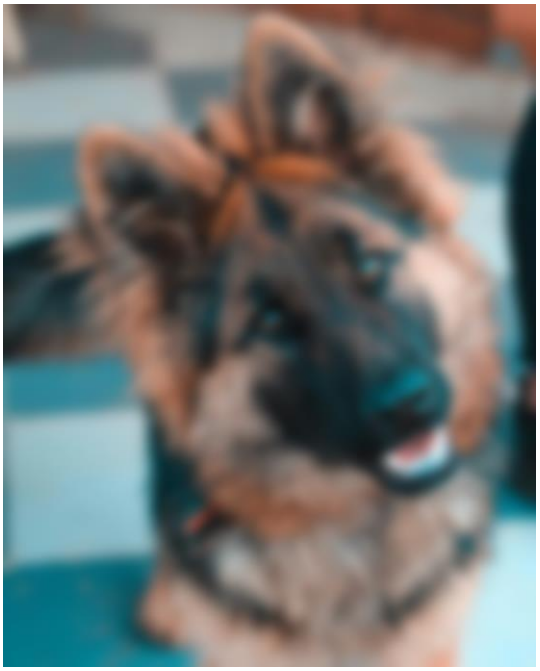


Image 2:

Image 3: 1_Limniu 2021_1_2021_06_02-13-08-37-958.PNG

Task 7: Image operations

Image 1. Dog.jpg

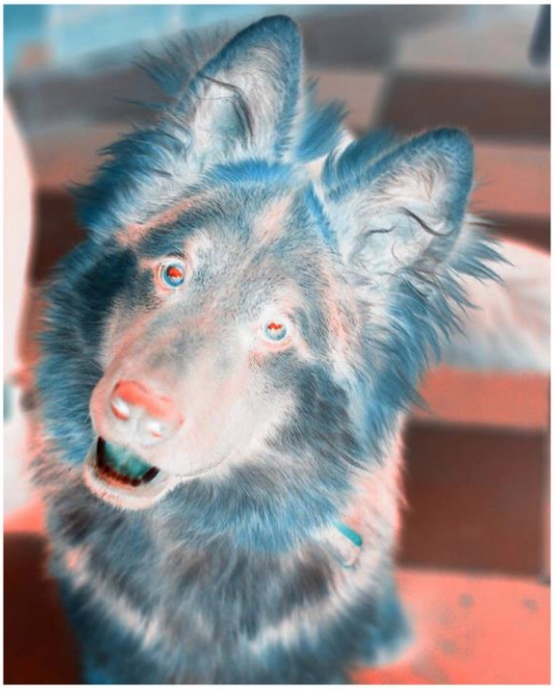


Image 2: 1_Limniu 2021_1_2021_06_02-13-08-37-941.PNG

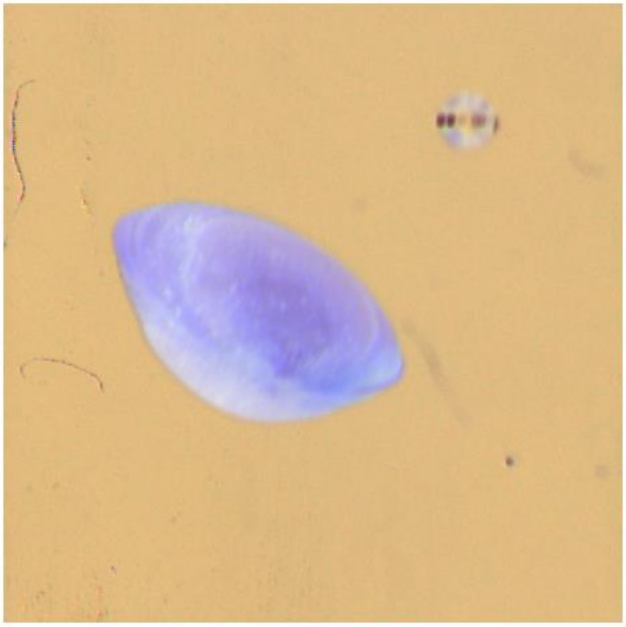



Image 3: 2_Sphaer 2021_1_2021_06_03-11-22-54-894.PNG